Conditional Neural Bayes Ratio Estimation (cNBRE)

Sam Leeney

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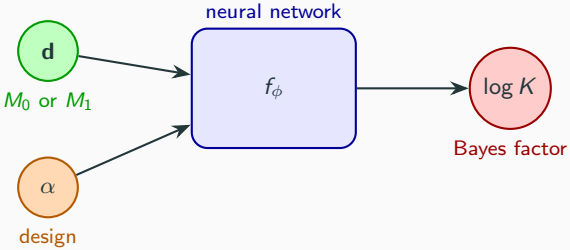
cNBRE: a classifier that outputs $\log K(\mathbf{d} \mid \alpha)$

- Simulate training data under M_1 and M_0 across design space α.
- Train $f_\phi(\mathbf{d}, \alpha)$ with exponential loss

$$\ell = \mathbb{E}_{M_1}\left[e^{-f_\phi/2}\right] + \mathbb{E}_{M_0}\left[e^{+f_\phi/2}\right].$$

- Optimum: $f_\phi^*(\mathbf{d}, \alpha) = \log K(\mathbf{d} \mid \alpha)$ (with balanced classes).
- Cost: train once (4–10 A100 GPU-hours), then evaluate in ms for any α.



Data $\mathbf{d}$ (simulated under M_0 or M_1) and design parameters α fed into neural network; optimised f_ϕ^* predicts $\log K$.

REACH: detection vs observation time

Forward models

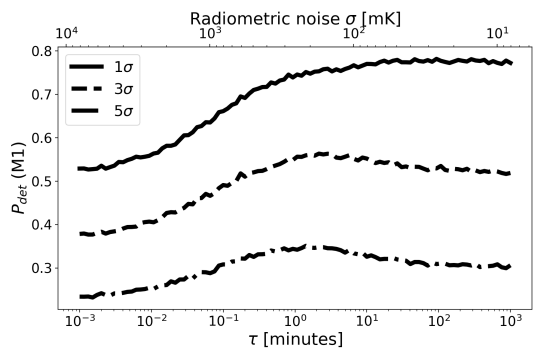
$$M_1 : \ \mathbf{d} = \mathbf{d}_{\rm fg} + \mathbf{d}_{21} + \mathbf{n}$$
$$M_0 : \ \mathbf{d} = \mathbf{d}_{\rm fg} + \mathbf{n}$$

Design parameters $\alpha = (\tau, \psi)$.

Detection probability

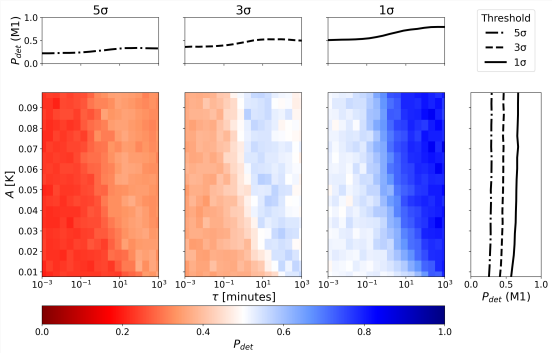
$$P_{\rm det}(\alpha) = \Pr_{M_1,\alpha}\left[\log K(\mathbf{d} \mid \alpha) > \log K_{\rm crit}\right].$$

- $\log K_{\rm crit}$: σ-equivalent FPR under M_0.
- $P_{\rm det}$ plateaus after $\sim 10\,\min$ due to a **systematics noise floor**.

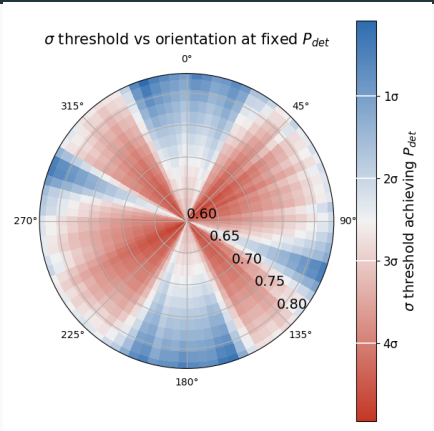


$P_{\rm det}$ vs. observation time τ at σ-equivalent thresholds (1σ, 3σ, 5σ).

Detection landscapes



P_{det} over signal amplitude and observation time.

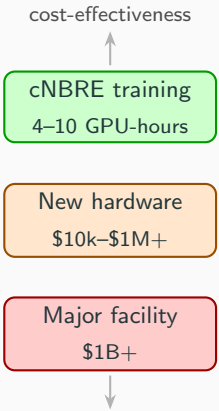


$\sim$20pp variation in P_{det} with orientation (3σ, τ=10 h).

Four-fold symmetry from dipole beam geometry.

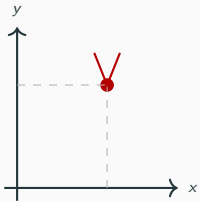
A note on cost

- GPUs are not free — but physical experiments cost *far* more for comparable sensitivity gains.
- cNBRE training: **4–10 A100 GPU-hours** (one-time).
- Physical alternatives for similar ΔP_{det}:
  - New antenna design & fabrication
  - Lower-loss cables & connectors
  - Additional calibration hardware
  - Extended observation campaigns
- SKA: ∼€2 B. LHC: ∼$9 B.
- ∼20pp gain from antenna orientation costs **nothing** once identified.



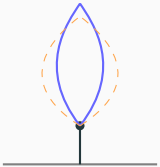
Next steps

Instrument optimisation



Antenna location

$\alpha = (x, y)$



Antenna geometry

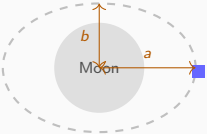
$\alpha =$ beam params

Calibration hardware

$\alpha = \ell$ (cable length)

Uncertainty propagation

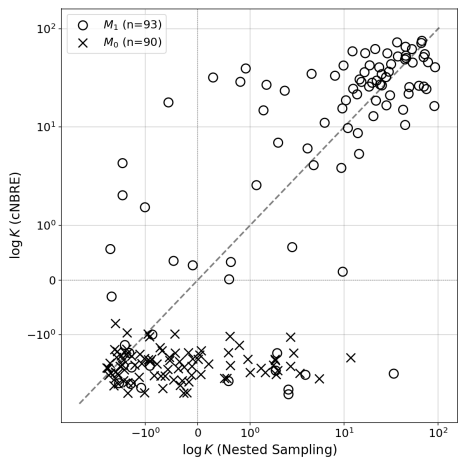
$\alpha = \delta_{\mathrm{VNA}}, \delta_{\mathrm{LNA}}, \ldots$



Lunar orbit location

$\alpha = (a, b, i, \ldots)$

Null testing



cNBRE vs. nested sampling $\log K$ on held-out simulations.

- f_ϕ broadly tracks NS $\log K$ for M_1 (signal) data
- For M_0 (null), f_ϕ is strongly negative, conservative null tests
- With our NS settings, cNBRE produced fewer false positives
- Train once (4–10 A100 GPU-h); inference: ms vs. hours per NS run

Discussion

Null testing

- f_ϕ broadly tracks NS $\log K$ for M_1 data
- For M_0, f_ϕ strongly negative, conservative null tests
- Fewer false positives than NS at comparable compute
- Train once (4–10 GPU-h); inference in ms

Instrument optimisation

- f_ϕ smooth in $\alpha \Rightarrow$ gradient-based optimisation
- Antenna location ($\alpha = x, y$), geometry, orientation
- Calibration hardware ($\alpha = \ell$)
- Component tolerance ($\alpha = \delta_{\mathrm{VNA}}, \ldots$) $\rightarrow P_{\mathrm{det}}$
- Lunar orbit location ($\alpha = a, b, i, \ldots$)